

1. Use a Truth Table to prove that $A \downarrow B$ is logically equivalent to $\neg(A \vee B)$:

A	B	$A \downarrow B$	$\neg(A \vee B)$	$(A \downarrow B) \Leftrightarrow (\neg(A \vee B))$
T	T	F	F	T
T	F	F	F	T
F	T	F	F	T
F	F	T	T	T

2. Use a Truth Table to prove that $A \downarrow A$ is logically equivalent to $\neg A$:

A	B	$A \downarrow A$	$\neg A$	$(A \downarrow B) \Leftrightarrow \neg A$
T	T	F	F	T
T	F	F	F	T
F	T	T	T	T
F	F	T	T	T

3. Use a Truth Table to prove that $(A \downarrow B) \downarrow (A \downarrow B)$ is logically equivalent to $A \vee B$:

A	B	$(A \downarrow B)$	$(A \downarrow B) \downarrow (A \downarrow B)$	$A \vee B$	$(A \downarrow B) \downarrow (A \downarrow B) \Leftrightarrow A \vee B$
T	T	F	T	T	T
T	F	F	T	T	T
F	T	F	T	T	T
F	F	T	F	F	T

4. Use a Truth Table to prove that $(A \downarrow A) \downarrow (B \downarrow B)$ is logically equivalent to $A \wedge B$:

A	B	$(A \downarrow A)$	$(B \downarrow B)$	$(A \downarrow A) \downarrow (B \downarrow B)$	$A \wedge B$	$(A \downarrow A) \downarrow (B \downarrow B) \Leftrightarrow A \wedge B$
T	T	F	F	T	T	T
T	F	F	T	F	F	T
F	T	T	F	F	F	T
F	F	T	T	F	F	T

5. Use a Truth Table to prove that $A \Rightarrow B$ is logically equivalent to $\neg A \vee B$:

A	B	$A \Rightarrow B$	$\neg A$	$\neg A \vee B$	$A \Rightarrow B \Leftrightarrow \neg A \vee B$
T	T	T	F	T	T
T	F	F	F	F	T
F	T	T	T	T	T
F	F	T	T	T	T

6. Use what you have learned above to come up with an expression using just A, B, and the NOR operator that is logically equivalent to $A \Rightarrow B$.

$$((A \downarrow A) \downarrow B) \downarrow ((A \downarrow A) \downarrow B)$$

A	B	$A \Rightarrow B$	$(A \downarrow A)$	$(A \downarrow A) \downarrow B$	$((A \downarrow A) \downarrow B) \downarrow ((A \downarrow A) \downarrow B)$	$A \Rightarrow B \Leftrightarrow ((A \downarrow A) \downarrow B) \downarrow ((A \downarrow A) \downarrow B)$
T	T	T	F	F	T	T
T	F	F	F	T	F	T
F	T	T	T	F	T	T
F	F	T	T	F	T	T

7. Prove the following argument by Deduction. $A \wedge C, B \wedge X \therefore (A \wedge B) \vee Q$

1	$A \wedge C$	
2	$B \wedge X$	
3	A	$\wedge E 1$
4	B	$\wedge E 2$
5	$A \wedge B$	$\wedge I 3, 4$
6	$(A \wedge B) \vee Q$	$\vee I 5$

8. Prove the following argument by Deduction. $A \therefore B \Rightarrow (A \vee C)$

Construct a proof for the argument: $A \therefore B \rightarrow (A \vee C)$

1		A			
2			B		
3				$(A \vee C)$	$\vee I 1$
4		$B \rightarrow (A \vee C)$	$\rightarrow I 2-3$		

 NEW LINE

 NEW SUBPROOF

😊 Congratulations! This proof is correct.

9. Prove the following argument by Deduction. $(P \wedge Q) \wedge R \therefore (P \vee S) \wedge (R \vee S)$

1		$(P \wedge Q) \wedge R$	
2		$(P \wedge Q)$	$\wedge E 1$
3		P	$\wedge E 2$
4		R	$\wedge E 1$
5		$(P \vee S)$	$\vee I 3$
6		$(R \vee S)$	$\vee I 4$
7		$(P \vee S) \wedge (R \vee S)$	$\wedge I 5, 6$

 NEW LINE

 NEW SUBPROOF

😊 Congratulations! This proof is correct.

CHECK PROOF

START OVER

10. Prove the following argument by Deduction. $Y \Rightarrow (X \Rightarrow Z) \therefore (Y \wedge X) \Rightarrow Z$

1	$Y \rightarrow (X \rightarrow Z)$	
2	$Y \wedge X$	
3	Y	$\wedge E\ 2$
4	$X \rightarrow Z$	$\rightarrow E\ 3, 1$
5	X	$\wedge E\ 2$
6	Z	$\rightarrow E\ 5, 4$
7	$(Y \wedge X) \rightarrow Z$	$\rightarrow I\ 2-6$

 NEW LINE

 NEW SUBPROOF

😊 Congratulations! This proof is correct.